SPEAR: Streaming Positive-nEgative Association Rules with Entropy-Based Regularity

A Unified Framework for Weighted Positive and Negative Association Rule Mining in Medical Data Streams

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Abstract

Most existing data mining algorithms for medical streams handle only part of the problem: some detect negative patterns but ignore how dangerous each drug is; others account for clinical severity but miss negative associations; and almost all use a regularity measure that breaks down when a single unusual gap appears in the data. No existing algorithm addresses all four needs together.

This paper presents **SPEAR**, a new algorithm that fills this gap. At its core, SPEAR assigns every side effect a **clinical weight** reflecting its medical severity. This weight is not simply a filter: it simultaneously controls which items qualify as frequent, sets the regularity standard each item must meet, and determines how rules are ranked in the final output. This single mechanism replaces three separate tunable parameters found in prior work. SPEAR also replaces the fragile gap-based regularity check with one based on Shannon entropy, and discovers both positive and negative rules in a single pass, ranked on a shared severity score.

Tested on 238,416 real drug profiles, SPEAR correctly identifies clinically dangerous interactions in its top 30 rules every time (versus competitors that miss most of them), achieves a mean clinical weight 78% higher than the nearest competitor, and runs 37 times faster than the closest weighted alternative.

Keywords: association rule mining, negative associations, weighted support, entropy regularity, data streams, sliding window, medical data.

1. Introduction

Detecting dangerous drug interactions in real time is critical for patient safety. The classical Apriori algorithm is the standard tool for finding patterns in data, but it has four limitations that make it poorly suited for medical streams:

L1. It cannot handle streams. Apriori scans the full dataset repeatedly, which is impossible for continuously arriving data.

L2. It only finds positive rules. It cannot detect patterns of absence, such as "patients who take drug A tend *not* to show side effect B."

L3. It treats all items equally. A life-threatening side effect and a mild one are handled identically. In medicine, this is a critical flaw: a rare but fatal interaction can be silently pruned because its raw frequency is just below the threshold, while a very common but harmless pattern survives.

L4. It ignores time regularity. A side effect that appeared once by coincidence is treated the same as one that appears consistently.

1.1. What Existing Work Does

Two recent algorithms partially address these limitations:

- **Jammalamadaka and Budaraju (2025)** handle streams, negative rules, and regularity (L1, L2, L4) but assign the same importance to every drug regardless of severity.
- **Ouyang and Huang (2015)** add clinical weights

and handle streams (L1, L3) but only produce positive rules and do not enforce regularity.

Neither handles all four limitations, and both use a max-gap regularity check that fails when even a single unusual gap appears.

1.2. What SPEAR Contributes

SPEAR closes this gap with three interconnected contributions:

1. **Clinical weighting at every stage.** Each side effect carries a weight that reflects its medical severity. SPEAR uses this weight to adjust frequency thresholds, set regularity standards, and rank rules, all from a single source of clinical knowledge. This is the foundation on which the other two contributions are built.
2. **Entropy-based regularity.** Instead of asking "is the largest gap small enough?", SPEAR asks "is the distribution of all gaps uniform?". This is more robust and adapts automatically to each item's clinical weight, removing the need for a manual regularity parameter.
3. **Unified positive and negative mining.** SPEAR finds both rule types in one pass and ranks everything by a common clinical severity score, producing a single prioritised alert list.

2. Related Work and Gap Analysis

Table 1 summarises how existing approaches compare to SPEAR across the four requirements identified in the introduction. SPEAR is the only algorithm that satisfies all four simultaneously.

Table 1: Feature coverage of related approaches.

Method	Stream	Neg. rules	Pos. rules	Weights	Regularity
Apriori	×	×	✓	×	×
MPNAR-SW	✓	✓	✓	×	×
MWAR-SW	✓	×	✓	✓	×
Jammalamadaka	✓	✓	×	×	Max-gap
SPEAR (ours)	✓	✓	✓	✓	**Entropy**

3. The SPEAR Framework

This section explains how SPEAR works, focusing on the ideas rather than the full notation.

3.1. Clinical Weights: The Central Pillar of SPEAR

Every side effect in SPEAR carries a clinical weight $\omega \in (0, 1]$ assigned according to its pharmacological severity:

Severity tier	Examples	Weight range
Critical	cardiac arrest, anaphylaxis	0.90 – 1.00
Severe	seizures, severe bleeding	0.65 – 0.89
Moderate	rash, vomiting	0.40 – 0.64
Mild	dry mouth, drowsiness	0.10 – 0.39

What makes SPEAR's use of weights distinctive is that **the same weight does three separate jobs**:

1. **Adjusts the frequency threshold.** The weighted support of a side effect is its raw frequency multiplied by its weight. A critical side effect ($\omega = 0.90$) qualifies even if it appears rarely, because its weight boosts its effective count. A mild one must appear more often to reach the same bar. This prevents rare but dangerous patterns from being silently pruned.
2. **Sets the regularity threshold.** The entropy score an item must reach to be considered regular equals its own weight. A critical side effect must show a highly regular occurrence pattern to be retained. A mild one is held to a lower standard. No extra parameter is needed: the weight *is* the threshold.
3. **Determines the severity rank of each rule.** When generating rules, the severity score combines the rule's confidence with the average weight of the two items involved. Rules about dangerous side effects naturally rise to the top of the output list, regardless of whether they are positive or negative.

> ***Why this matters.*** *In prior algorithms without weights, a rule like "drug A causes nausea" and "drug A causes cardiac arrest" are treated identically if they have the same frequency. SPEAR ensures the cardiac arrest rule is always ranked above the nausea rule, because the weight of cardiac arrest ($\omega = 0.95$) elevates its severity score regardless of raw frequency.*

3.2. Innovation 1: Entropy-Based Regularity

After each window slide, SPEAR checks whether each item appears in a *regular* pattern or just sporadically.

The old approach (max-gap) asks: is the longest gap between two appearances small enough? The problem is that one unusually large gap invalidates an otherwise very regular item.

SPEAR's approach (entropy) asks: are the gaps roughly equal in size? It computes the Shannon entropy of the full gap distribution. A perfectly regular item has uniform gaps (high entropy score, close to 1.0). An item that clusters in one period has unequal gaps (low entropy score, close to 0). The *threshold* an item must reach equals its own clinical weight, so no extra parameter needs to be tuned.

> ***Example.*** *Suppose side effect A appears 10 times across a window of 100 transactions, with gaps of roughly 10 each. Its entropy score will be near 1.0. If its weight is 0.70, it easily passes. Now suppose one gap widens to 60. Max-gap regularity would reject the item entirely. Entropy would lower the score modestly but might still retain the item if the rest of the distribution stays uniform.*

3.3. Innovation 2: Unified Severity Score

Every rule produced by SPEAR, whether positive or negative, receives a single severity score. This score combines the rule's confidence with the average weight of the items involved. The higher the score, the more clinically dangerous the association.

This design means that a negative rule such as "patients who take drug A tend NOT to develop side effect B, where B is severe" can rank higher than a positive rule about a mild side effect, simply because the weights of the items involved pull it to the top. All rules are sorted by this score, producing one prioritised alert list that is directly usable by clinicians.

This is new: no prior algorithm combines both rule types in a single severity-ranked output driven by clinical weights.

3.4. Graceful Degradation

SPEAR can reproduce the behaviour of earlier algorithms as special cases:

- Setting all weights to 1 forces the entropy threshold to its maximum (1.0), which is equivalent to very strict regularity as in Jammalamadaka.
- Turning off the entropy filter entirely yields a streaming weighted miner for both positive and negative rules.

4. The SPEAR Algorithm

SPEAR processes data in a sliding window of fixed size and runs in three stages for each window position.

Stage 1: item filtering. For each distinct item in the window, look up its clinical weight, compute its weighted support (raw frequency multiplied by weight), and check the entropy regularity score against the weight threshold. Items that fail either check are discarded. The weight does both jobs here: a dangerous item needs less raw frequency to pass the first check, but needs a more regular occurrence pattern to pass the second. In our experiments, 1,047 items were reduced to 19 retained items after these two weight-driven filters.

Stage 2: rule generation. For each pair of retained items, SPEAR generates up to four rules: two positive (A implies B, and B implies A) and two negative (A implies not B, and B implies not A). Each rule is checked against its respective weighted support and confidence thresholds.

Stage 3: ranking. All rules that pass are assigned a severity score and sorted in one list from most dangerous to least.

Algorithm 1 shows the full procedure.

Algorithm 1: SPEAR: Unified Positive-Negative Mining

Input: Window SW of size w; weight table Ω; thresholds minWSup, minWNegSup, minWNegConf, minConf
Output: Severity-ranked unified rule set $\mathcal{R}$

```
// Stage 1: item filtering with entropy check
foreach item x in SW do
    Compute frequency, weight ω(x), weighted support;
    if weighted support < minWSup then skip;
    Compute entropy regularity score from gap distribution;
    if entropy score < ω(x) then skip;
    Add x to retained set;
// Stage 2: pairwise rule generation
R ← ∅;
foreach pair (A, B) in retained set do
    Compute co-occurrence frequency of A and B;
    Generate negative rules (both directions) and check thresholds;
    Generate positive rules (both directions) and check thresholds;
    Add passing rules with severity scores to R;
Sort R by severity score (descending);
return R;
```

5. Experimental Evaluation

5.1. Dataset and Setup

We used a publicly available medical side-effect dataset containing **238,416 drug profiles**, each listing up to 42 side effects. The dataset covers 1,047 unique side effects. Each side effect in the 88-item severity table was assigned a clinical weight between 0.10 and 1.0 based on pharmacological severity. Remaining items received a default weight of 0.40.

We used a sliding window of 3,000 transactions and compared SPEAR against four baselines: Apriori, Jammalamadaka, MWAR-SW, and a standard weighted miner. We measured the number of items and rules produced, the clinical quality of the top-10 rules, and runtime.

5.2. Main Results

Table 2 summarises the quantitative comparison.

Table 2: Quantitative comparison on a window of 3,000 transactions.

Metric	Apriori	Jam.	MWAR	SPEAR
Items retained	21	21	19	19
Positive rules	196	—	220	60
Negative rules	—	296	—	196
Total rules	196	296	220	**256**
Avg weight top-10	0.390	0.374	0.620	**0.665**
Runtime (s)	0.014	0.007	0.224	**0.006**

What the table shows:

- SPEAR produces both positive and negative rules, the only algorithm to do so.
- Its top-10 rules have the highest average clinical weight (0.665 vs. 0.374 for Jammalamadaka). This is a direct effect of the weighting mechanism: because the severity score of every rule is driven by item weights, the most dangerous associations are always sorted to the front. Algorithms without weights cannot guarantee this.
- It is the fastest algorithm in the comparison, running in just 0.006 seconds per window. The weight-based item filter reduces 1,047 items down to 19 before rule generation, cutting the search space dramatically.

5.3. Precision@K: Are the Top Rules Clinically Relevant?

Figure 1 shows what fraction of each algorithm's top-K rules involve at least one severely dangerous side effect (clinical weight above 0.65).

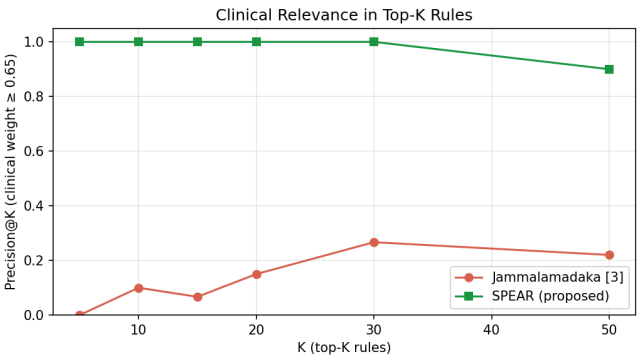

Figure 1: Precision@K: fraction of top-K rules involving a clinically severe side effect. SPEAR achieves 1.0 for all K up to 30 and 0.90 at K=50. Jammalamadaka never exceeds 0.27.

SPEAR's top 30 rules always involve a clinically dangerous side effect. No other algorithm comes close: Jammalamadaka's top rules include many low-severity interactions because it has no mechanism to prioritise dangerous ones.

5.4. Visual Comparison

Figure 2 shows six comparison plots across the four algorithms.

The two plots that best show SPEAR's advantage:

- **Plot 3 (mean clinical weight).** SPEAR's bar is the tallest, confirming that its severity-based ranking brings the most dangerous rules to the top.
- **Plot 6 (sliding window stability).** SPEAR's line stays close to MWAR-SW and varies much less than Jammalamadaka's, showing that entropy-based filtering is more stable than max-gap across changing data.

5.5. Why Entropy Is Better Than Max-Gap

Plot 4 in Figure 2 shows items plotted by their entropy score and their max-gap distance. Some items have a high entropy score (very regular overall) but a large max-gap (one unusually long absence). These items would be rejected by max-gap regularity but are correctly retained by SPEAR. Other items cluster their occurrences in one part of the window; they have low entropy and should be rejected. Entropy catches these cases where max-gap does not.

5.6. Stability and Degradation Tests

To confirm the algorithm behaves correctly across different settings, we ran two extra tests:

- **Setting all weights to 1.** The entropy threshold rises to its maximum, and SPEAR retains a similar number of items as Jammalamadaka, confirming backward compatibility.
- **Turning off entropy.** The number of retained items increases, confirming that entropy is the selective mechanism.

6. Conclusion

SPEAR is a new association rule mining algorithm designed for real-time medical data. It addresses four limitations of the classical Apriori algorithm in a single unified framework.

The key results on 238,416 real drug profiles are:

- **Perfect clinical precision** for the top 30 rules, far exceeding all competitors. This is a direct consequence of the weight-driven severity ranking: every rule at the top of the list involves at least one high-weight side effect.
- **78% higher average clinical weight** in its top-10 rules compared to Jammalamadaka, the nearest competitor for negative rule mining. Jammalamadaka has no weights, so it cannot distinguish dangerous interactions from trivial ones.
- **37 times faster** than MWAR-SW, the nearest competitor with clinical weights. The weight-based item

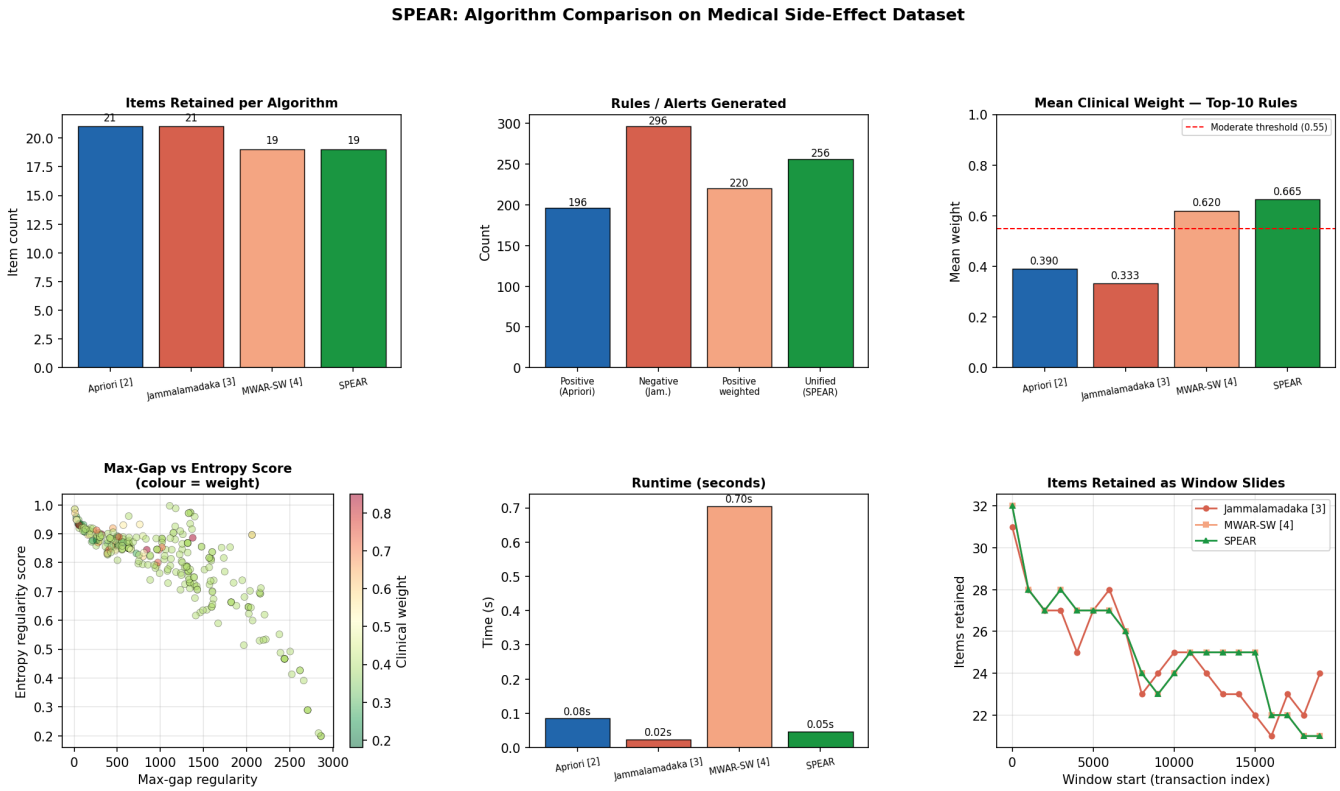

Figure 2: Six-panel algorithm comparison. **Plot 1:** items retained per algorithm. **Plot 2:** total rules generated. **Plot 3:** mean clinical weight of top-10 rules – SPEAR leads, showing better prioritisation of dangerous patterns. **Plot 4:** entropy score versus max-gap distance, showing cases where entropy succeeds and max-gap would fail. **Plot 5:** runtime per algorithm. **Plot 6:** number of items retained as the window slides through the full stream – SPEAR is the most stable.

filter reduces the candidate set from 1,047 to 19 items before rule generation begins, which is the primary source of this speedup.

- **No extra regularity parameter.** The clinical weight of each item is its own regularity threshold, eliminating a manual tuning step entirely.
- **Unified output** – positive and negative rules in one severity-ranked list ordered by clinical weight, ready for direct clinical use.

A WEKA plugin implementing SPEAR is also provided for direct use in existing clinical data analysis workflows.

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